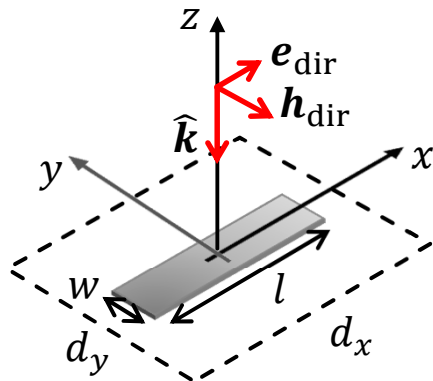


Problem 2

Calculate the radiation patterns from the FSS in the figure below, under plane-wave illumination. Use a windowing technique with 10×10 unit cells.

Dipole strips in free space



As an example, consider the following geometric parameters

```
dx = 20*mm;    theta = 0;  
dy = 20*mm;    phi   = 0;  
w  = 1 *mm;    TM incidence  
l  = 15*mm;  
Freq = 10e9;
```

- Plot the active element pattern as a function of θ .
- At what angle of incidence θ_{in} is an higher order Floquet mode entering the visible region, as predicted by theory?
- Plot the total patterns for normal incidence $\theta = 0^\circ$ and for oblique incidence $\theta = \theta_{\text{in}}$
- How high is the level of the grating lobe for oblique incidence $\theta = \theta_{\text{in}}$ compared to the main lobe and why?